Sharon Basovich

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Education

University of Waterloo Expected 2029
BCS | Honours Computer Science, Co-op *Waterloo, ON*

Internships

Software Engineering | *RBC* | Toronto, ON Summer 2026

- Developed an internal engineering intelligence platform used by **200+ banking apps across 1,200 repositories** daily
- Built an automated service-health analysis pipeline ingesting SonarQube, DAST, GitHub, and Snyk data through GraphQL/REST APIs into MongoDB, saving service owners **12,000+ hours annually**
- Developed a modular React dashboard in an Nx monorepo to monitor service health and vulnerabilities
- Added and maintained automated tests with Vitest and Playwright to validate app logic and reduce regressions

Software Automation Engineering | *Magna International* | Newmarket, ON Summer 2024

- Programmed surface-geometry, force, and leverage computations and visualizations using VBA, Excel, and CATIA V5 (CAD) to generate safety-compliance reports, **automating 50+ hours** of previously manual analysis
- Streamlined ingestion of 100+ materials records and 78 engineering drawings into Siemens Teamcenter PLM
- Soldered wiring-harness prototypes and tested automotive hood-latch system failures in validation lab

Projects

UWaterloo High Performance Computing | *Sponsored by Citadel & Jane Street* | Waterloo, Ontario Present

- Developing XPU, a C++23 interface for CPU/CUDA memory management, math, data layouts, and execution
- Building a C++ variational Monte Carlo engine for homogeneous electron-gas simulations using a Slater–Jastrow trial wavefunction and Metropolis sampling

Marble Investments | *$3M AUM Investment Fund* | Waterloo, ON Present

- Developed a React research graph persisting 500+ companies and 1,500+ relationships for the investment team
- Created a news pipeline monitoring **2,000 articles daily**, with LLM extraction of relevance and impact to portfolio
- Built a Python pair screener evaluating 1M+ stock pairs to shortlist roughly 50–100 candidates for analyst review

SkyCell | *1st Place at Apex International Hackathon* | Boston, MA Apr. 2025 – Jun. 2025

- Launched a custom 915 MHz LoRa/Meshtastic telemetry link **30 km into space** to establish a range of 70 km
- Built Python software to parse NMEA GPS data, integrate environmental sensors, and recover radio connections
- Automated RTL-SDR spectrum sweeps from 30 MHz to 1.5 GHz at 2.4 MS/s, capturing I/Q data for RF analysis

Lions Robotics | *Vice President & Lead Programmer* | Richmond Hill, ON Jun. 2024 – Jun. 2026

- Programmed autonomous robot PID, odometry, and Monte Carlo localization in C++, placing **10th in Canada**
- Implemented concurrent C++ tasks using sensor feedback to coordinate scoring and motion mechanisms
- Led a 56-member organization to six World Championship qualifications while managing a $140k annual budget

Contests & Awards

Canadian Computing Competition Honor Roll | *Placed Top 2.5% in Senior Division*

Carnegie Mellon University picoCTF Cybersecurity Competition | *Ranked 113th out of 10,460 (Top 1%)*

VEX Robotics World Championship Judges Award | *1 of 10 Recipients Among 9,565 Teams*

11-Time Hackathon Winner | *Hack the North, General Catalyst, Ramp, etc.*

Technical Skills

Languages: C++, C, Python, TypeScript, Java, GraphQL, SQL, VBA
Technologies: Git, Docker, Linux, React, MongoDB, Node.js, Vitest, Playwright, PostgreSQL, OpenCV, PyTorch
Interests: Horology, Science Fiction, MMA, Poker, Skiing, Cooking, Mechanical Keyboards